Experiment Name: Configuring static routing using CLI command.

Objectives

The main objective of this experiment is to create two different computer networks using two routers and two switches. Another objective is to configure the routers properly using CLI commands. The final objective is to ensure successful data and message transfer between PCs located in separate networks using static routing.

Introduction

In this experiment, static routing is configured using Command Line Interface (CLI) commands to establish communication between two different networks. Each network consists of two PCs connected through a switch and a router. The purpose of this experiment is to understand how routers can be manually configured using CLI commands to forward data packets between separate networks. After completing the configuration, successful communication between all PCs in both networks should be achieved.

Required Devices:

- **Two Cisco 1841 routers**
- **Two Cisco 2960 switches**
- **Six PCs**
- **Straight-through cables for LAN links**

Procedures

Step 1: **Placing all required devices**

At first, four PCs, two switches, and two routers were placed on the Cisco Packet Tracer workspace. Each switch was positioned near a router, and two PCs were placed near each switch so that they could form two separate networks.

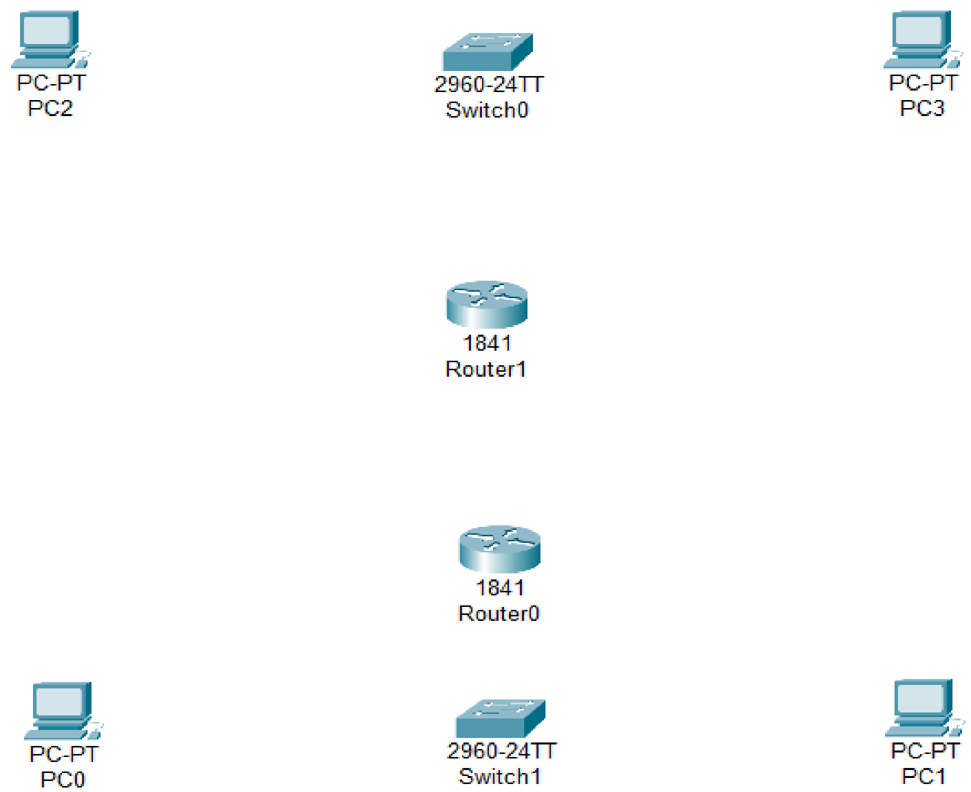

Figure 7.1: Placing all the devices in the workstation

Step 2: Connecting the devices

The PCs were connected to their respective switches, and the switches were connected to their respective routers. In both cases, copper straight-through cables were used to establish proper physical connections..

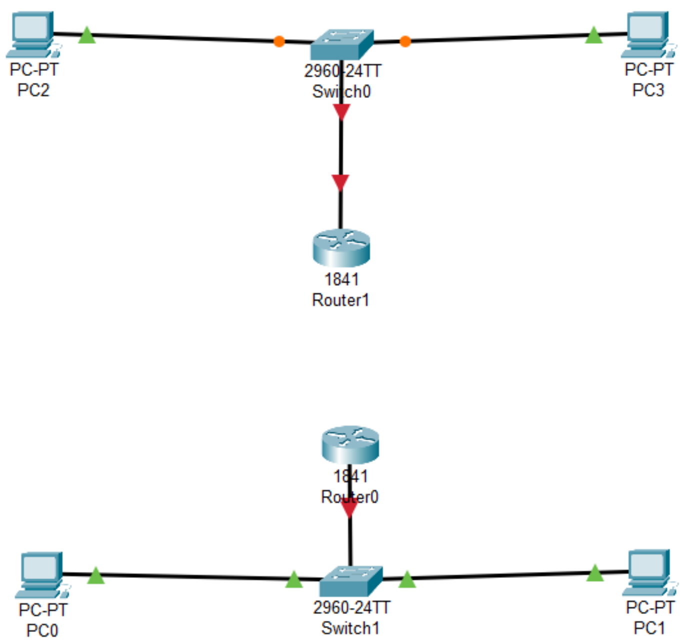

Figure 7.2: Device Connection

Step 3: Adding serial ports to routers

Router0 was powered off from the physical tab, and a WIC-2T module was inserted into the router. The router was then powered on again. The same process was repeated for Router1 to enable serial communication on both routers.

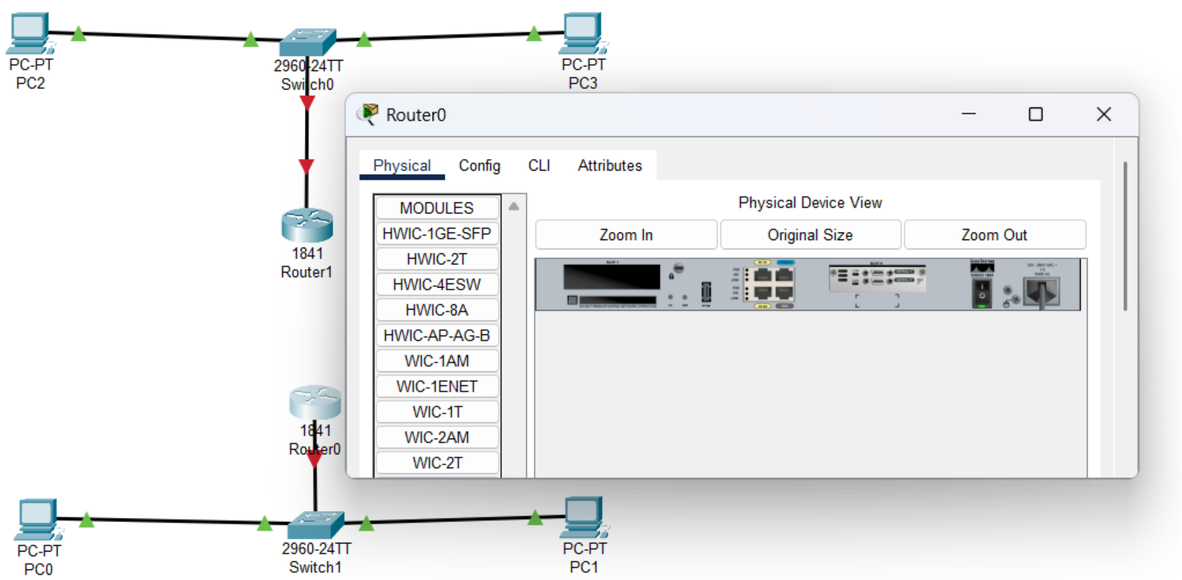

Figure 7.4: Adding Serial ports to routers

Step 4: Connecting Routers using Serial DTE Cable

A serial DTE cable was then used to connect Router0's Serial0/0/0 interface with Router1's Serial0/0/0 interface. This created the WAN link between the two routers.

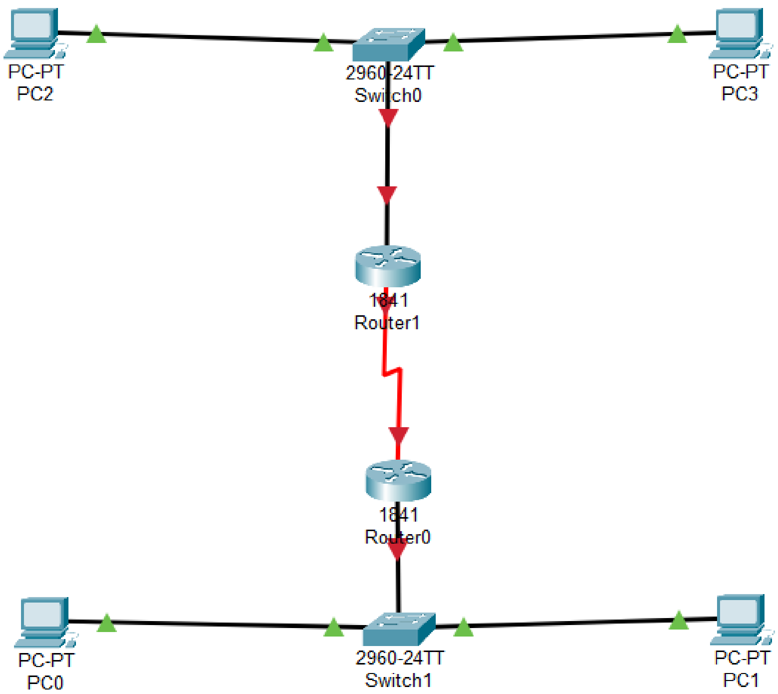

Figure 7.4: Connecting Routers via Serial DTE Cable

Step 5: Assigning IP Address to PCs

Each PC was assigned an IP address and default gateway according to its LAN. The gateway corresponds to the FastEthernet0/0 interface of the connected router.

PC IP Address - Gateway

- PC0 192.168.10.2 - 192.168.10.1
- PC1 192.168.10.3 - 192.168.10.1
- PC2 192.168.11.2 - 192.168.11.1
- PC3 192.168.11.3 - 192.168.11.1

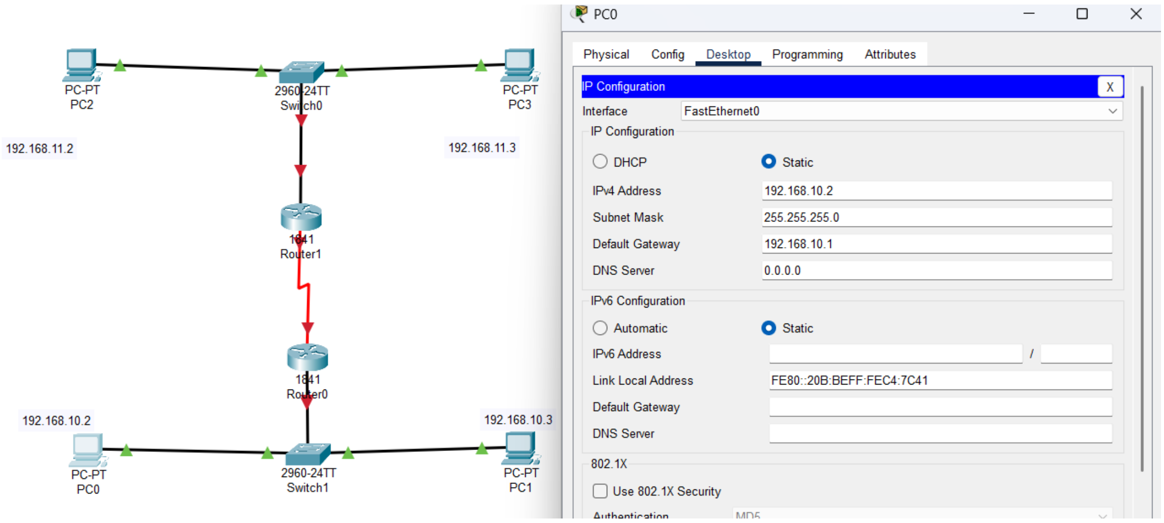

Figure 7.5: Configuring IP Adress for all PCs

Step 6: Configuring Router0 using CLI Commands

Router0 was accessed through the CLI option. The router was enabled using CLI commands. Then FastEthernet0/0 and Serial0/0/0 interfaces were configured with suitable IP addresses. After that, a static route was configured using the IP route command. This configuration was carefully noted because it would be required while configuring Router1.

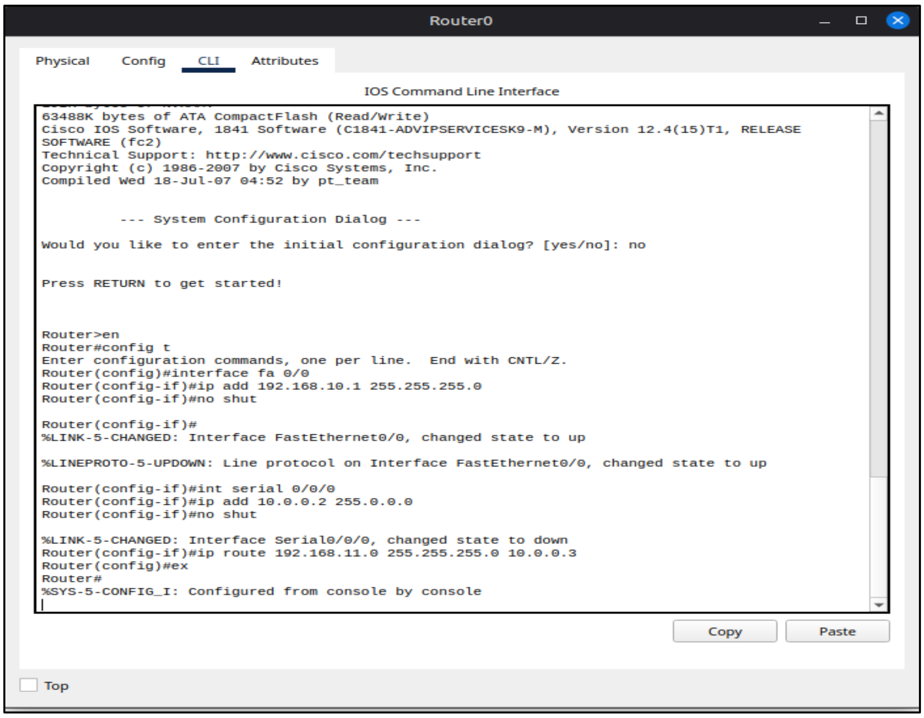

Figure 7.6: Configuring Router0 using CLI commands

Step 7: Configuring Router0 using CLI Commands

After completing the configuration of Router0, Router1 was accessed through the CLI option. The router was enabled, and FastEthernet0/0 and Serial0/0/0 interfaces were configured using CLI commands. Then the static route was added according to the configuration of Router0 so that both routers could exchange data properly.

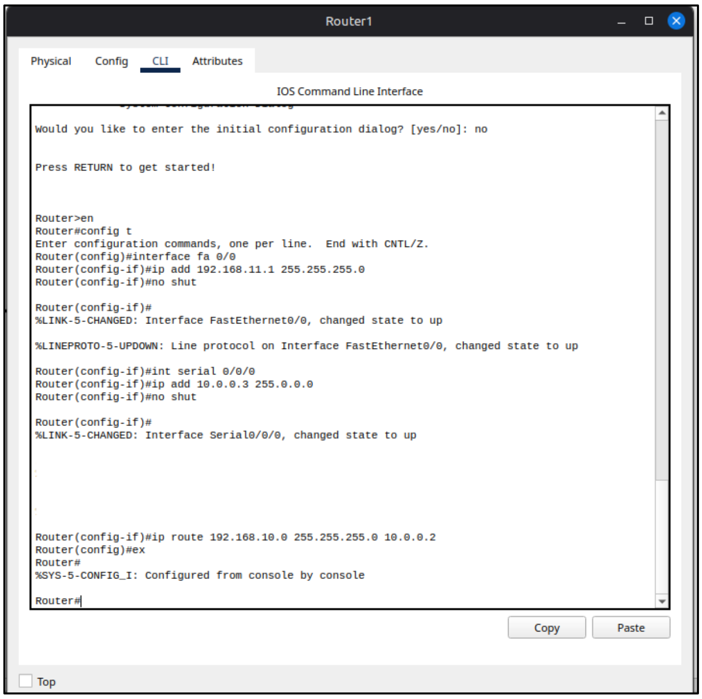

Figure 7.7: Configuring Router1 using CLI commands

Step 8: Testing Connectivity

Finally, message transfer was tested between the PCs of both networks. Several message transfers were performed between different combinations of PCs. All test results were successful, proving that the static routing configuration was correct.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC3	ICMP		0.000	N	1	(edit)	
	Successful	PC1	PC2	ICMP		0.000	N	2	(edit)	
	Successful	PC3	PC1	ICMP		0.000	N	3	(edit)	
	Successful	PC2	PC1	ICMP		0.000	N	4	(edit)	

Figure 7.8: Successful Message after Testing

Result

The result of this experiment was successful. Messages were transferred from any one PC to any other PC within the same network as well as across different networks. Every possible combination of message transfer between the four PCs worked properly without any packet loss.

Conclusion

In this experiment, two separate networks were successfully connected using static routing configured through CLI commands. The use of WIC-2T modules, serial connections, and correct IP addressing played an important role in achieving successful connectivity. This experiment helped in understanding the practical implementation of static routing and the importance of proper router configuration for reliable network communication.